CMPE460L / SWEN360L

Software Design and Engineering Laboratory

BatchBite: Preorder and Production Platform for Small Food Businesses

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1. Project Overview and Objectives

1.1 Project Overview

BatchBite is a management tool that will be used for making orders and handling preproduction. This platform is supposed to make the management of food product listings, customer orders, production batches, order pickups, and order tracking more convenient. Small food businesses often experience problems in order management, availability tracking, preparation, and order pickup. BatchBite solves these problems by offering a unified platform, in which customers can view available products, order products, choose pickup time slots, and track order status. There are four types of users in this platform: Customers, Business Owners, Kitchen Staff, and Administrators. Business Owners can manage products, create production batches, set up pickup slots, and observe orders. Kitchen Staff can see the preparation requirements and update order status. Customers can view products, order products, and view order history. Administrators can manage user accounts and permissions. The first phase of development includes requirements gathering, needs analysis, and preparation of design artifacts for future phases.

1.2 Project Objectives

The main objectives of BatchBite is:

- To serving as a hub for managing the food products and preorders made by customers.

- To allow the Business Owner to create food product listings and batch production scheduling.

- To avoid over-sold items by managing available quantities, production capacity, and pickup slots.

- To enable customers to place orders and choose pickup slots, while tracking their orders.

- To help the Kitchen Staff in organizing their work and providing customer customization and packaging details.

- To maintain accurate orders and enabling proper order status transition.

- To provide role-based user access to Customers, Business Owners, Kitchen Staff, and Administrator.

- To establish the initial system design for future development and testing.

1.3 Scope of Phase 1

Phase 1 concentrates on requirements analysis and system design. At this stage, it is essential to understand the needs and the features of the system that should be developed. Thus, we create user stories, acceptance criteria, wireframes, and UML diagrams. We also conduct sprint planning, develop a GitHub repository, and create a Scrum board. This phase lays the

groundwork for implementation in the coming development phases. It does not denote the completion of the fully implemented application.

2. Team information and roles

Team member	Role	Responsibilities
Shaikha	Scrum Master	Organizes Scrum meetings, manages the Kanban board, tracks progress and blockers, and documents meetings and sprint activities.
Nehad	UI/UX Designer + Diagrams	Designs application screens and user flows; creates wireframes, UML diagrams, and the system architecture diagram.
Aaya	Mobile Frontend Developer	Plans and implements the interface, navigation, customer screens, forms, buttons, and interactions.
Zain	Backend + Database Developer	Designs data structures for users, products, orders, batches, authentication, and storage.
Malaak	Backend + Integration Developer	Defines business logic, order processing, capacity rules, production logic, and frontend/backend integration.
Marwa	Testing + QA Developer	Prepares test scenarios, validates requirements, identifies bugs, and documents testing and fixes.

3. Sprint Planning

Sprint Duration: October 1 – October 18

Sprint Goal: Sprint 1 aims to lay the groundwork for the development of the BatchBite system through setting up the functional and non-functional requirements of the system, creating the user stories for the entire group of users, as well as the creation of the design artifacts of the system, such as the wireframes, UML diagrams, and the architecture of the system. In addition, the GitHub repository will be set up.

Task Breakdown:

- Functional Requirements: Malaak
- Non-Functional Requirements: Aaya
- User Stories (Customer & Administrator): Aaya
- User Stories (Business Owner & Kitchen Staff): Malaak
- Wireframes: Marwa
- UML Diagrams (Use Case, Class, Sequence): Nehad & Zain
- Activity Diagrams: Marwa
- System Architecture Diagram: Nehad & Zain
- Documentation & Meeting Logs: Shaikha
- GitHub Repository & Kanban Board Management: Shaikha
- Sprint Planning & Task Tracking: Shaikha
- Sprint Retrospective: Shaikha & All Team Members

4. Functional Requirements

This section describes actions that BatchBite shall do. The requirements are categorized based on modules and are assigned to a priority level (High, Medium, and Low), along with the sprint during which they will be implemented. Requirements for Sprint 1 make up the primary core requirements; others are added now to incorporate them into the design.

Accounts and Access

ID	Requirement	Priority	Sprint
FR-1	The system will provide role-based account options for Customers, Business Owners, Kitchen Staff, and Administrators.	High	1
FR-2	The system will provide options for secure registration and login, and will limit access to all screens and functions based on the role permitted to access them.	High	1
FR-3	The system will provide Administrators with the ability to manage user accounts and set or modify user roles.	Medium	1

Products and Production Batches

ID	Requirement	Priority	Sprint
FR-4	The system shall provide Business Owners with the ability to create and modify listings of their products with pictures, descriptions, variants, ingredients information, pricing, and available quantity.	High	1

FR-5	The system shall provide Business Owners with the ability to create batches for product manufacturing with an order deadline, maximum quantity, and pick-up date.	High	1
FR-6	The system shall automatically close orders for a particular batch upon reaching the deadline or maximum quantity.	High	1

Cart, Checkout, and Stock Reservation

ID	Requirement	Priority	Sprint
FR-7	The system shall allow Customers to browse open batches and place products in their cart.	High	1
FR-8	The system shall show the order summary (Items, quantities, total, pickup slot) before the customer confirms the checkout.	High	1
FR-9	The system shall confirm a payment through a simulated payment confirmation.	High	1
FR-10	The system shall reserve the stock when the checkout occurs by holding it, in a way that the stock will not be accessible to another Customer while the hold is active	High	1
FR-11	The system shall cancel a reservation of the stock and the capacity when a hold expires or when an order is cancelled.	High	1

Pickup Slots

ID	Requirement	Priority	Sprint
FR-12	The system shall enable the Business Owners to set pickup timeslots with maximum orders per time slot.	High	1
FR-13	The system shall enable Customers to choose a suitable pickup time slot and prevent them from selecting time slots that are full.	High	1

Order Tracking and Production Workflow

ID	Requirement	Priority	Sprint

FR-14	The system shall monitor each order throughout the process of Confirmed, Preparing, Ready, and Collected.	High	1
FR-15	The system shall only permit valid status changes (from Confirmed to Preparing to Ready to Collected) and will prevent invalid status changes such as marking an order as Collected before its status becomes Ready.	High	1
FR-16	The system shall enable customers to access their order history and status of each order.	High	1
FR-17	The system shall provide a Business Owner Dashboard to view orders, cancellation of orders, and progress of production.	High	1
FR-18	The system shall enable the Kitchen Staff to access orders for the current batch along with customization and packaging information.	High	1
FR-19	The system shall remove canceled orders from the active production board.	Medium	1
FR-20	The system shall inform Customers about order confirmation, order ready for pick up, and delay notifications and maintain a record of all the notifications made accessible to the staff.	Medium	1
FR-21	The system shall consider two successive actions of a particular action as one action and shall not create duplicate records.	Medium	1

Order Changes and Cancellations

ID	Requirement	Priority	Sprint
FR-22	The system shall allow the Customers to modify quantity/variant prior to the cutoff provided only if there is sufficient stock and production capacity, recalculate the totals and the simulated refund but retain the same history of the order.	Medium	2
FR-23	The system shall maintain the recipe version and price version that were used for each confirmed order to ensure that future edits cannot affect the already confirmed orders.	Medium	2

Ingredients, Collection, and Reporting

ID	Requirement	Priority	Sprint
FR-24	The system shall derive the required ingredients based on confirmed orders, scaling the recipes with the quantity of the order, compare them against available and reserved stock, and create a purchasing list for any shortfall.	Medium	2
FR-25	The system shall create a unique QR code for each order and only confirm the delivery of the order when it is Ready.	Medium	2
FR-26	The system shall organize the products into preparation batches on the production board retaining all customer-specific notes.	Low	2
FR-27	The system shall report the amount of orders confirmed/cancelled/produced/collected by batch and product.	Low	3

5. Non-Functional Requirements

This section discusses the quality attributes and constraints that BatchBite must satisfy to offer a reliable, secure and user-friendly experience to its customers, business owners, kitchen staff and administrators.

ID	Non-Functional Requirement	Priority
NFR-1	**Usability:** The system shall offer a clear and consistent interface that allows users to explore the features offered depending on their roles.	High
NFR-2	**Performance:** The system shall display product lists, order details, and order status information in not more than 3 seconds under normal operating conditions. .	High
NFR-3	**Security:** The system shall restrict access to certain pages based on user roles and authentication is needed for protected functions.	High
NFR-4	**Data Integrity:** The system shall ensure accurate product quantities, production capacity, pickup-slot availability, and order records when placing or canceling orders.	High
NFR-5	**Reliability:** The system shall prevent invalid order-status transition and duplicate orders as a result of multiple submissions.	High

NFR-6	**Compatibility:** The application shall offer a consistent experience when accessed using popular web browsers and mobile platforms.	Medium
NFR-7	**Maintainability:** The system shall have a modular architecture, and the functionalities can be changed independently without necessarily impacting other functions such as the product list, checkout and order tracking.	Medium
NFR-8	**Availability of Information:** Customers shall be able to access their order status and order history while authorized staff shall be able to access relevant information required for their roles.	High

These requirements will guide the design and implementation of BatchBite throughout the development sprints.

6. User Stories and Acceptance Criteria

The Business Owner is the owner of a small food business who schedules production batches, creates/manages orders, and tracks their progress. The stories below include acceptance criteria and functional requirements.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
BO-1	As a Business Owner, I would like to be able to create product listings with photos, variants, ingredients and pricing, in order to present the exact offerings to the customers.	Product save occurs for all fields and the product appears in the customer catalogue; errors appear for missing required fields.	FR-4	High	1
BO-2	As a Business Owner, I would like to be able to create production batches with deadlines, max quantities and collection dates, in order to avoid overselling.	Batch creation includes all three values; orders close automatically either at the set deadline or when the maximum quantity is reached.	FR-5, FR-6	High	1
BO-3	As a Business Owner, I would like to be able to create pickup slots with order limits, in order to	When the slot is full, the field is indicated as such and cannot be chosen by customers.	FR-12, FR-13	High	1

	avoid overcrowding during collection periods.				
BO-4	As a Business Owner, I would like to be able to see all the orders in a particular batch in a dashboard, in order to get my confirmed demand.	Dashboard displays every order with information on customer, ordered items, current status and assigned pick-up slot along with the total amount of ordered products compared to maximum quantity.	FR-17	High	1
BO-5	As a Business Owner, I would like to be able to record any order cancellations, in order to release stock and capacity.	Cancelled order is not included in the active orders anymore; its quantity and pick-up slot go back into batch.	FR-11, FR-17, FR-19	Medium	1
BO-6	As a Business Owner, I would like to be able to update the production progress in order to keep order status updated for my customers.	Status update is reflected in customer view along with notification; invalid statuses are not accepted.	FR-14, FR-15, FR-20	Medium	1
BO-7	As a Business Owner, I would like to be able to approve or deny the change requests of customers before the cut-off, in order to avoid exceeding my stock and capacity.	Order update can occur only if the stock and capacity are sufficient enough; total amount and the refunded price are updated and previous order history is preserved.	FR-22	Medium	2
BO-8	As a Business Owner, I would like to be able to have ingredient	Shopping list indicates required amount of ingredients compared to currently available amount.	FR-24	Medium	2

	requirements calculated based on confirmed orders.				
BO-9	As a Business Owner, I would like to have ordered, cancelled, produced and collected quantities in a report so that I can plan better future batches	Report can be filtered based on batch and product and compares planned with actual amount.	FR-27	Low	3

5. User Stories: Kitchen Staff

Kitchen Staff prepares and delivers orders. They need to be able to have a clear visibility of their tasks and to be able to track their progress and confirm delivery.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
KS-1	As Kitchen Staff, I need to see all the orders which are currently prepared, so I will know what should be prepared.	Order list contains the items, their variations, amount of products, and time slots for picking up of the chosen batch.	FR-18	High	1
KS-2	As Kitchen Staff, I need to see the customizations and packing details, so nothing special would be forgotten.	Customer's notes appear on each order in the production process.	FR-18	High	1
KS-3	As Kitchen Staff, I need to be able to set the order state as Preparing and Ready, so the progress could be tracked.	Only transition states of Confirmed to Preparing and Preparing to Ready are possible, and the customer receives notification about readiness of the order.	FR-14, FR-15, FR-20	High	1
KS-4	As Kitchen Staff, I need to be able to confirm the delivery of an order only in	Collected state is disabled for orders which are not Ready and message appears about it.	FR-15	High	1

	case it is Ready, so wrong handovers will not happen.				
KS-5	As Kitchen Staff, I need cancelled orders to vanish from the production panel, so I will not prepare unnecessary orders.	Cancelled orders are invisible in the active board, but remain in the history of orders.	FR-19	Medium	1
KS-6	As Kitchen Staff, I need the ability to send the status update once even if I tapped twice, so I will not create duplicates.	Duplicate submission of the same update results in single status change and single history update.	FR-21	Medium	1
KS-7	As Kitchen Staff, I need similar products to be grouped into preparation batches, so I could work effectively.	Orders in the board are classified depending on product and variation and include notes of each customer.	FR-26	Low	2
KS-8	As Kitchen Staff, I need the ability to scan a QR code of the customer upon delivery so collection is fast and traceable.	Valid code of Ready order turns it into Collected state; invalid or used code will be rejected.	FR-25	Medium	2

Order Status Flow:
Confirmed-> Preparing->Ready->Collected. A cancelled order exits the active flow and keeps its history. No status may be skipped.

6. User Stories: Customer
The Customer uses BatchBite to browse available products, place and manage orders, select pickup times, and stay informed about order progress.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
CU-1	As a Customer, I want to browse through	The system displays the available products along with	FR-4, FR-5,	High	1

	available products and batches of production in order to see the available food products and select the one I want to order.	their pictures, description, variants, price, and available quantity. The customer can see open batches and cannot make an order in closed batches.	FR-6, FR-7		
CU-2	As a Customer, I want to add items to my cart and see my order summary, so that I can check my selected products, quantities, total, and pickup time slot before placing the order.	Customers can add products to the cart and set their quantities. The system shows the selected products, quantities, total and pickup time slot before placing the order.	FR-7, FR-8	High	1
CU-3	As a Customer, I would like to place an order and get payment confirmation, so that I will know that my order has been successfully placed.	The system reserves the required stock during placing an order and shows the simulated payment confirmation. The order is confirmed when the checkout process is completed. In case the checkout fails or there is no required stock, then no order is created.	FR-9, FR-10, FR-11	High	1
CU-4	As a Customer, I would like to select an available pickup time slot, so that I can pick up my order in a convenient time period.	The system shows available pickup time slots for the selected batch. Customers can choose time slots that have available capacity and cannot choose the ones that are full. The selected time slot is added to the order summary before confirming the checkout.	FR-8, FR-12, FR-13	High	1
CU-5	As a Customer, I would like to track the status of my order and see my order history, so that I will know the information about my orders and previous purchases.	The system shows the customer's orders and their statuses. Order statuses are following each other according to the order: Confirmed, Preparing, Ready, Collected. Customers can access their previous orders and see the latest status of each order.	FR-14, FR-15, FR-16	High	1

CU-6	As a Customer, I would like to get notifications about my orders, so that I will know when my order is confirmed, ready to pick up, or delayed.	The system sends the customers notifications when their orders are confirmed and ready to pick up. Also, when the order is delayed, the system sends the delay notification. The system saves all notifications, which the authorized staff can access.	FR-20	Medium	1
CU-7	As a Customer, I would like to update my order before the cutoff time, so that I can change my quantities and product variants if I need.	Customers can update their quantities and product variants before the cutoff time. The system checks the availability of stock and capacity before accepting the update and calculating the new order total. When the total changes, the system updates the simulated refund accordingly if necessary. The system saves the history of updates.	FR-22	Medium	2
CU-8	As a Customer, I would like to cancel my order when cancellation is allowed, so that I can cancel order I no longer need.	When cancellation is allowed, the system registers the cancellation, frees the reserved stock and capacity, and cancels the order from the active production board. The cancellation is reflected in the order records	FR-11, FR-17, FR-19	Medium	1

7. User Stories: Administrator

The Administrator manages user accounts and user roles to ensure that access to BatchBite is properly controlled.

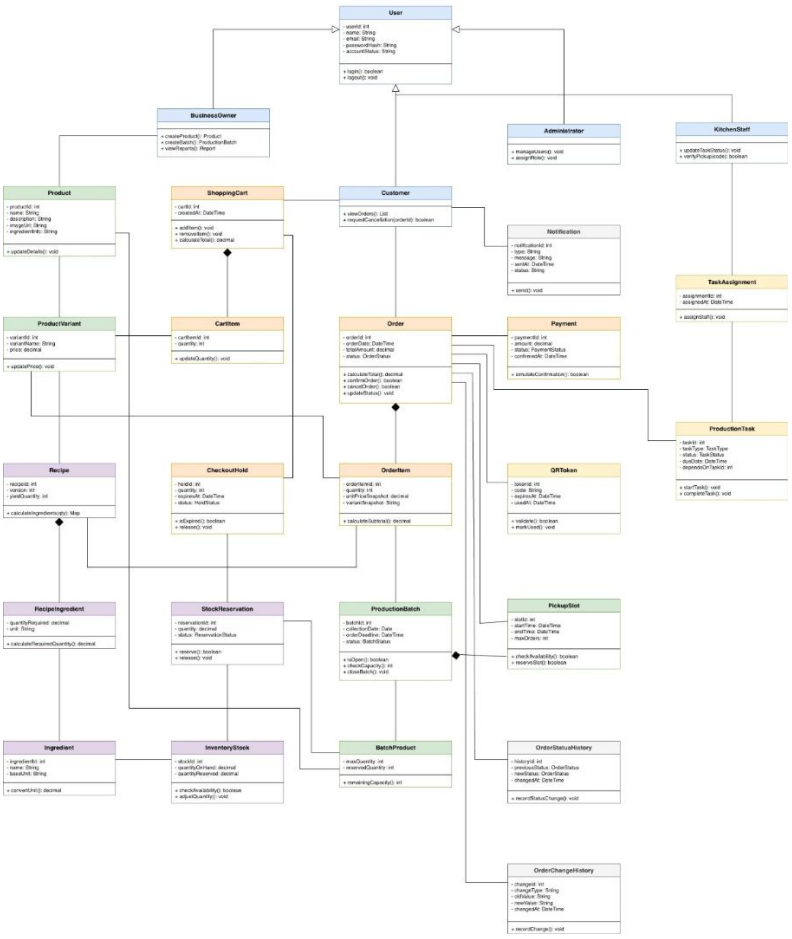
ID	**User Story**	**Acceptance Criteria**	**Related FR**	**Priority**	**Sprint**
AD-1	As an Administrator, I would like to manage user accounts, so that I	The Administrator can manage user accounts, and	FR-3	Medium	1

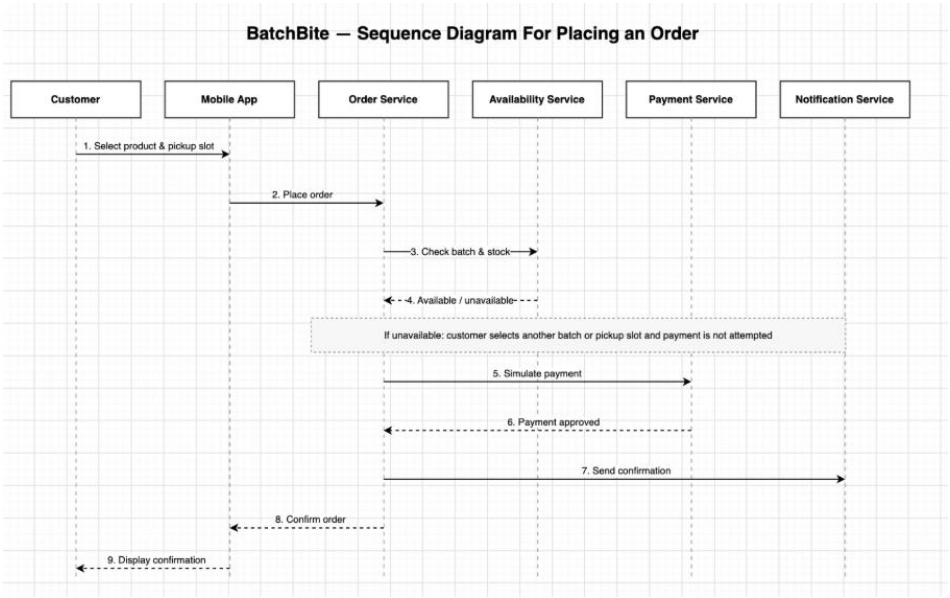
	can maintain user access to BatchBite.	changes are reflected in the system.			
AD-2	As an Administrator, I would like to assign and modify user roles, so that each user has the appropriate level of access.	The Administrator can assign or modify roles for Customers, Business Owners, and Kitchen Staff. The system applies the updated permissions according to the assigned role.	FR-1, FR-2, FR-3	Medium	1

7. Initial System Design

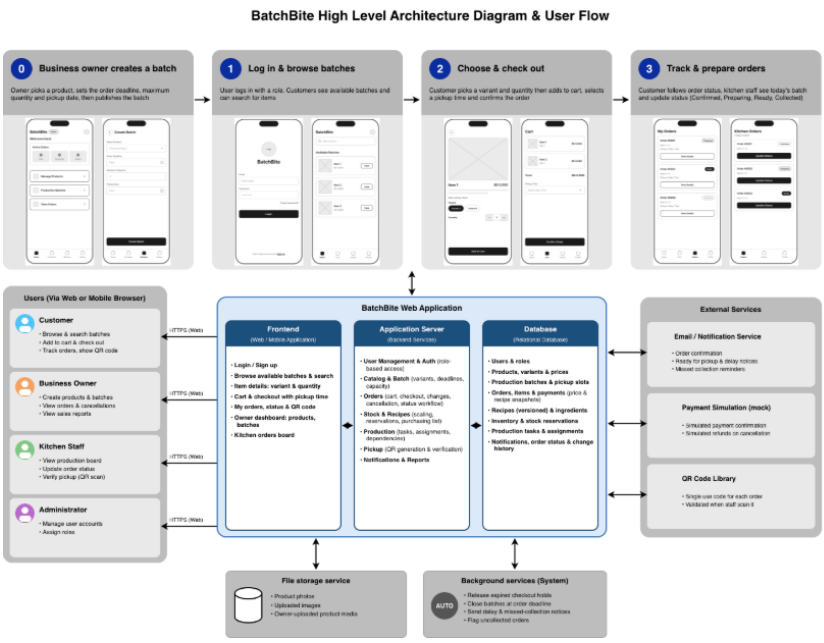


UML Class Diagram - Batch Bites





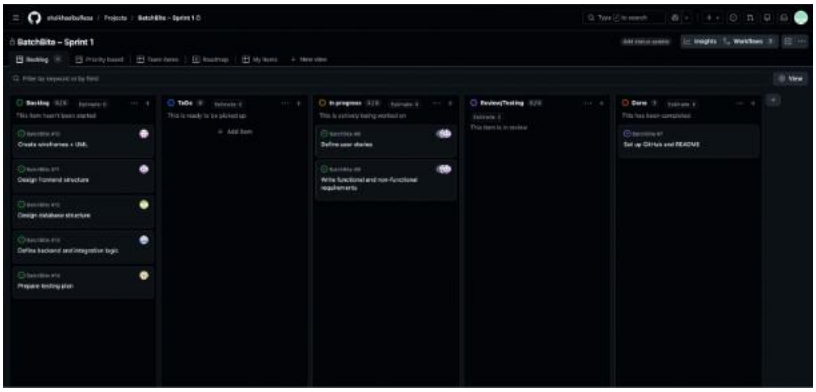
8. System Architecture



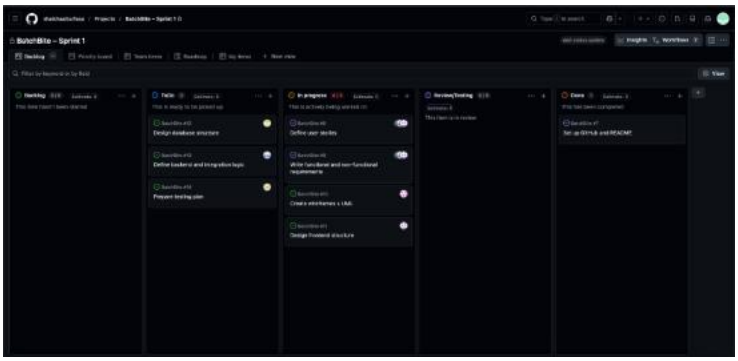
9. GitHub Repository and Scrum Board

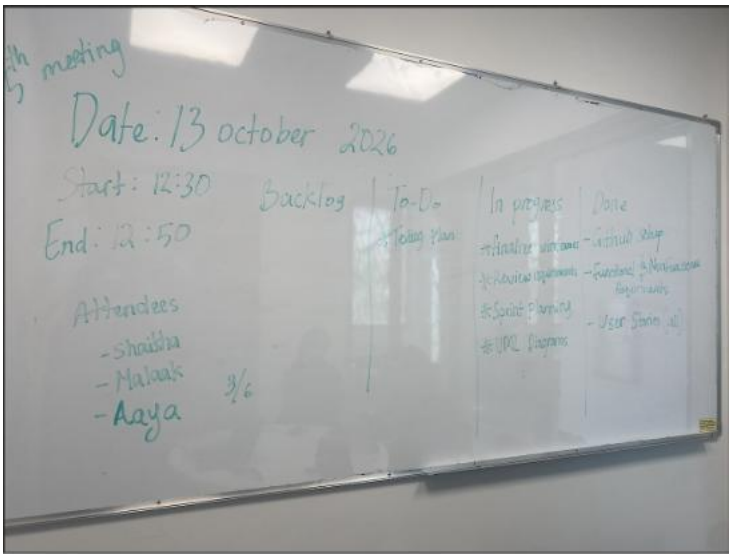
GitHub Link: https://github.com/shaikhaalbuflasa/BatchBite

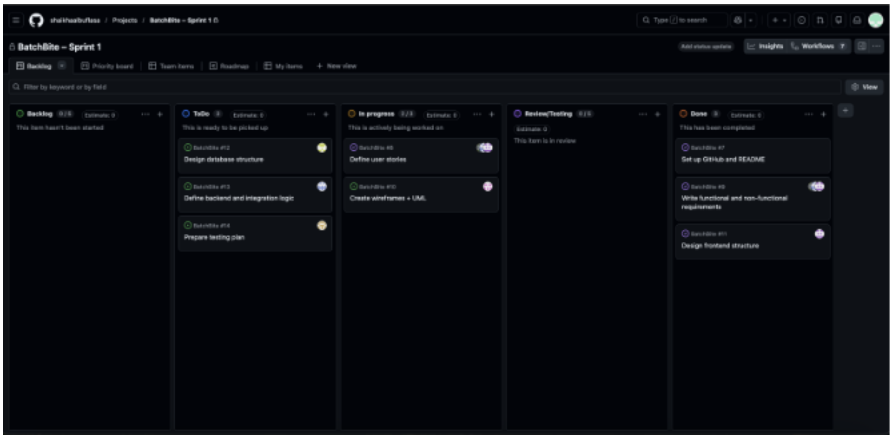
Beginning (October 6): Initial state of the BatchBite Kanban board on October 6, 2026, with tasks organized by status to track the team's progress throughout Sprint 1.



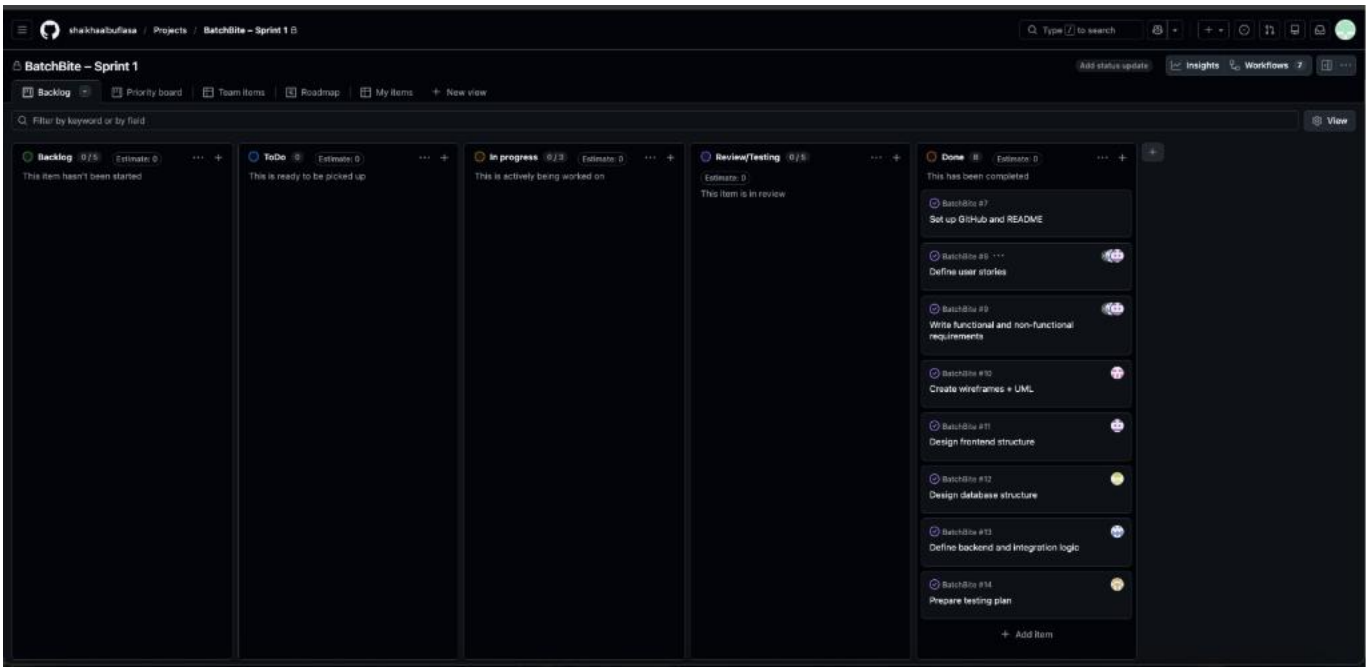
Mid-Sprint (October 8): Multiple tasks in progress







End (October 18): tasks completed in ‘Done’


shakhaabuflasa / Projects / BatchBite – Sprint 1
BatchBite – Sprint 1
Backlog
Priority board
Team items
Roadmap
My items
New view
Filter by keyword or by field
View
Backlog 0 / 5
This item hasn't been started
ToDo 0
This is ready to be picked up
In progress 0 / 3
This is actively being worked on
Review/Testing 0 / 5
This item is in review
Done 8
This has been completed
Set up GitHub and README
Define user stories
Write functional and non-functional requirements
Create wireframes + UML
Design frontend structure
Design database structure
Define backend and integration logic
Prepare testing plan
Add item